Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for: **Neonatal Pain & Sucrose Guideline** Eastern and Northern Services	
Author of Clinical Guideline	Charlotte Woollacott, Senior Neonatal Nurse
Care Group/ Department responsible for Clinical Guideline	Women's and Children's Health / CWH/ Neonatal/ Trust-wide
Contact details	01271 322610/ Charlotte.woollacott@nhs.net
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SUMMARY

This guideline outlines the process to monitor and minimise pain for Neonates.

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1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 This document outlines guidelines for neonatal pain control including:

- Routine assessments for the detection of pain and stress using the ALPS-Neo Scale
- Reduction of number of painful procedures
- Strategies to prevent /reduce acute pain from invasive procedures

2. BACKGROUND

2.1 Neonates experience a greater sensitivity to pain (Anand et al 2006, Fitzgerald and Walker 2009) and can interpret excessive stimulation, handling and stress as pain (Warren and Bond, 2010)

2.2 Excessive or unresolved persistent pain or stress is damaging to the developing new born brain causing long term neurological impairment (Batton et al 2006)

2.3 Pain assessment in preterm infants is complex and difficult due to their limited capacity to express pain and stress, which is related to their physiological and neurobiological immaturity (Lundqvis et al, 2014).

2.4 All health professionals caring for Neonates have a professional and ethical responsibility to provide effective and safe pain management during uncomfortable and pain procedures (Harrison et al, 2010). Evidence demonstrates that controlling pain in the neonatal period is beneficial improving physiologic, behavioural and hormonal outcomes. (Witt et al 2016). Managing neonatal pain is important as repeated pain and stress can have a life-long impact on the infants themselves and on parental wellbeing (Valeri et al. 2015). In hospital, preterm infants can experience up to 12 painful procedures each day and less than half of these interventions use pain management (Cruz et al. 2016). Neonates can even experience normal touch as pain, due to immature neurones and increased density of nociceptive nerve endings in the skin, (Findlay 2004). Infants feel pain more severely than adults, (Sredl 2003), and exhibit greater cardiovascular, metabolic and hormonal responses to pain than children, (Anand 2001).

3. ROUTINE PAIN ASSESSMENT

3.1 Measurement of pain in neonates is complicated by the inability of neonates to verbalise pain and the subjective nature of assessment. Assessment of infants' pain and sedation levels will be made and documented routinely with observations and with procedures as required using 'the Astrid Lindgren and Lund Children's Hospitals Pain and Stress Assessment Scale for Preterm and Sick New born Infants' (ALPS-Neo) (Appendix 2) (Lundqvis et al, 2014) Appropriate management will be given.

3.2 With the ALPS–Neo scale, symptoms of pain and stress are monitored and registered simultaneously with other physiological parameters and this makes it possible to continuously evaluate pain and the need for analgesics or comfort measures as part of an overall clinical context (Lundqvis et al, 2014).

3.3 The ALPS-Neo consists of a five item scale and includes scores for:

- Facial expression
- Breathing pattern
- Tone of extremities
- Hand/ foot activity
- Level of activity

3.4 Every infant should be assessed for pain:

- On admission to NNU/ SCBU
- Before and after potentially painful interventions
- ITU/ HDU Monitored babies, at least every 4 hours or when vital signs are recorded
- Low Dependency Area. At least every 4 hours or when a care/intervention is done e.g. Nappy change, bath
- Transitional Care. Each time baby is seen or least 6 hourly
- At any point during a shift if the nurse or parent considers it necessary.
- Nurses can increase interval of pain assessment if required

3.5 Completing the ALPS-Neo assessment:

- Observe neonate for 15-30 seconds
- Score neonate for each parameter
- The total score is calculated automatically on the NNU vitals part of the chart on epic.
- If score indicates the need for intervention, refer to this guideline
- Reassess as guideline suggests

3.6 Teach parents to understand how pain and stress is assessed and how they can help to minimise it.

If parents are available they should be asked for their opinion, e.g. do they feel their baby is experiencing any pain?

See Appendix 2 for ALPS-Neo pain and stress scale and Appendix 3 for guidance.

4. PREVENTION

The prevention of pain in neonates should be the goal of all health care professionals who work with neonates. It is ethical duty but also because repeated painful exposures have the potential for deleterious consequences. Every healthcare facility caring for neonates should implement a pain assessment and strategies for minimising the number of painful procedures performed and management plan that includes routine assessment of pain, pharmacologic and non-pharmacologic therapies for the prevention of pain associated with routine procedures (Keels and Sethna 2016) Minimal handling (Witt et al 2016).

The most effective way to reduce neonatal pain is to reduce the number of procedures and infant handling by:

- Handling is the most direct source of stress for infants in NICU, (Reid and Freer 2000).
- Timing routine medical interventions with other procedures (e.g. daily physical examination with cares) but cluster care with caution (Holsti et al 2005)
- Evidence shows that stress levels increase with each activity undertaken; therefore, consideration should be taken into how many procedures are being carried out at any one time (Holsti et al 2005).
- Anticipate and group blood sampling to reduce frequency of invasive procedures
- Review the need for routine blood testing with the consultant on ward round
- If appropriate, use non-invasive monitoring such as transcutaneous or end-tidal CO2 monitoring to minimise blood sampling
- Attempts at cannulation should only be undertaken a maximum of 3 times before procedure is referred to senior team member

4.1 Non-pharmacological methods

Individualised supportive care intervention is always the primary alternative to optimise the infants' condition, as long as the infant does not have an obvious reason for pain (Lundqvis et al, 2014). See Appendix 4 which shows a checklist for optimising Neurodevelopmental Neonatal procedures.

4.2 Environmental

- Decrease lighting and noise level (Walden and Gibbins 2012)
- Comfortable and supportive surroundings
- Maintaining infant comfort ensuring baby is warm, has boundaries (following developmental care guidelines depending on gestation)

4.3 Non pharmacological procedures are more effective when used in combination

- Breastfeeding during procedures
- EBM with/without pacifier
- Sucrose with/without pacifier
- Containment holding
- Supportive positioning
- Swaddling
- Non-nutritive sucking
- Skin-to-skin/ kangaroo care (Pilai et al 2011)
- Carer/infant interaction
- Sensory stimulation – use of positive touch, massage, softly spoken maternal voice and maternal smell.

Maternal breast milk administered into the mouth has been shown to reduce procedural pain. If maternal breast milk is not available then small amounts of oral sucrose solution have also been shown to reduce procedural pain (NHSGGC Paediatrics 2022).

4.4 Procedural

- Treat anticipated procedure related pain prophylactically
- Minimal taping/remove tape gently and carefully. Use adhesive remover wipes where possible
- Optimise ventilation pre-procedure
- Consider using venepuncture not lancet blood sampling It is has been shown to be less painful for patients; however, there needs to be recognition that veins need to be preserved for cannulation (Shrestha & Adhikari 2012).

4.5 Pharmacological

- When individualised supportive care interventions do not have sufficient effect, the infant should have pharmacological treatment in order to minimise pain (Lundqvis et al, 2014)
- The multi-professional team must work to ensure appropriate pharmacological pain management
- Anticipate painful procedures and administer analgesia in advance, taking into account the time interval for the analgesia to take effect

The estimated time it takes for various routes of administration are given below:

- IV – within 10 minutes
- Oral sucrose - 2 minutes
- PO/PR – within 30 minutes
- Topical – within 1 hour
- Subcutaneous – within 5 minutes

4.6 Pharmacological options

- **Paracetamol** is affective for mild to moderate pain and also as an antipyretic during sepsis or post immunisations
- **Opiates** are most effective for moderate to severe pain (please see administration policy)
- **Sucrose**
- **Lignocaine**

Sedatives or muscle relaxants do not provide pain relief

5. NIDCAP 5 STEP DIALOGUE

5.1 The NICAP 5 step dialogue for all cares and procedures. The NIDCAP 5 step dialogue can form the framework in how to approach and support a baby in any interaction. Babies use their own touch and the touch of others to help calm themselves. Touch stimulates nerve endings in the skin, which leads to a release of hormones often referred to as 'love hormones' that produce pleasant sensations and feelings. Research has shown that the touch of a parent can help ill and preterm infants cope with painful procedures they receive as part of their care in neonatal intensive care units (NICUs).

1. Prepare - wash and warm hands to match baby's temperature. - Consider the environment as infants are more likely to feel uncomfortable with competing sensory inputs from a bright and loud environment.
2. Touch Permission - introduce yourself to the baby, by speaking softly to make your presence known. - Initiate touch using still hand/s on the infant over their body or cupped around their head
3. Tuning into the baby and pacing - observe the baby's behavioural cues, using a soft, slow voice to calm and settle.
4. Connecting to the baby - Giving the infant time to adjust and settle.
5. Breaking contact at the conclusion of touch experience - Withdraw hands very slowly, wait and watch.

(The Australasian NIDCAP Training Centre, 2018)

6. DOSAGE AND FREQUENCY OF SUCROSE

6.1 Oral sucrose is a non-pharmacological intervention that reduces pain responses in the newborn. Research has shown that it is effective in stable premature infants (>27weeks and usually several weeks old). However, it can be given at <27 weeks upon clinical decision.

Infant gestation	Dose
27+1 – 32 weeks gestation	0.1 – 0.5ml 24% Sucrose solution
32+1 weeks gestation	0.5 – 1ml 24% Sucrose solution

(NHSGGC Paediatrics 2022)

6.2 Sucrose is contraindicated in:

- Infants less than 27 weeks gestation.
- Infants being treated for NEC.
- If there are any neurological concerns affecting the suck, swallow or gag reflex.
- Critically ill neonates receiving appropriate intravenous analgesia and sedation who have low pain scores on handling.
- Neonates who are receiving muscle relaxant medication.
- Infants with a known fructose intolerance.

6.3 Guidance from the manufacturer says it can be administered up to 4 times within a 24 hour period in preterm infants, and up to 6-8 times within a 24 hour period in full term infants. It also states the vial can be used again by the same infant within a period of 8 hours, however best practice would be to dispose of the vial once finished with in clinical waste. If a further vial is needed open a new vial.

The manufacturer also states: Caution - Risk of chocking if the top tab section inadvertently dropped into the baby's mouth- dispose of the flat upper tab section carefully (Inspiration Health Care Group 2024)

6.4 Procedure:

- It should be prescribed on the baby's MAR.
- Discuss with parents/caregivers if available that you would like to offer the baby sucrose and the rationale for doing so.
- Wash hands/ use alcohol gel as per local unit policy.
- Ascertain whether sucrose is to be given with a pacifier (if parents have consented)
- Check Sucrose & patient identity as per unit protocol.
- Ensure baby is awake to reduce the risk of choking/aspiration, and positioned appropriately.
- Employ environmental and behavioural strategies to assist baby, utilise methods of swaddling, comfort holding, Kangaroo Care (skin to skin) or breastfeeding to potentiate the effectiveness of the sucrose and comfort the baby.
- Give the dose by either: dropping the prescribed dose onto the pacifier and gently place in their mouth or drop onto the tip of the baby's tongue gradually. Sucrose is ineffective if given via a naso/orogastric tube.

(East of England ODN 2024)

7. DOCUMENTATION AND EDUCATION

7.1 All aspects of pain management should be documented. Document pain assessment, type of intervention, effectiveness of treatment and family involvement.

7.2 All multi-professionals will receive education/training in the assessment and management of pain.

7.3 We should support parents by sharing information with them to increase parental knowledge of pain in infants and its management. Provide opportunities to discuss pain management with parents/carers. Guide and encourage parents on how they can touch and interact with their baby in the most effective way. Teach the family to understand their infant's pain and stress cues and what they can do to help relieve their symptoms.

8. MONITORING COMPLIANCE WITH THIS GUIDELINE

8.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per Royal Devon Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

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APPENDIX 1

SPECIFIC PAINFUL SITUATIONS AND MANAGEMENT

1. HEEL PRICK AND CANNULATION/ VENEPUNCTURE

- Pacifier + Breast milk (or sucrose if not available) 2 minutes prior to procedure
- Skin to skin kangaroo care
- Swaddling/Containment holding
- Breast feeding
- Parent/Carers holding infant

2. VITAMIN K / VACCINATION / CFAM NEEDLE INSERTION

- Pacifier + Breast milk (or sucrose if not available) 2 minutes prior to procedure
- Skin to skin kangaroo care
- Swaddling/Containment holding
- Post vaccination paracetamol

3. NASOGASTRIC / OROGASTRIC TUBE INSERTION

- Pacifier + Breast milk (or sucrose if not available) 2 minutes prior to procedure
- Breast milk
- Cuddling/Swaddling/Containment holding

4. CENTRAL LINES

- Consider bolus dose of Morphine prior to insertion of umbilical catheters for ventilated infants

5. CHEST DRAINS

Insertion

- Subcutaneous Lignocaine.
- Where the chest drain is not required as an emergency consider a bolus of Morphine 30 minutes prior to the procedure followed by a Morphine infusion.

Removal

- Consider a bolus of Morphine using the lowest recommended dose if self-ventilating. If this is not appropriate consider pacifier, sucrose and paracetamol.

6. ET INTUBATION

- ET intubation without the use of analgesia or sedation should be performed only for delivery room resuscitation or other life threatening situations or when there is no IV access.
- Follow local guidelines for ET intubation and ventilation.

7. ET SUCTION

- Minimal use of ET suctioning. Suction ET tubes when clinically indicated not as a routine procedure.
- Swaddling/Containment holding.

8. VENTILATED INFANTS

- Consider a continuous infusion of morphine sulphate. Adjust dose according to ALPS-Neo assessment score.
- Supportive positioning.
- Containment holding.
- Positive touch from parents.
- Skin to skin kangaroo care.

9. NON-SURGICAL CHRONIC PAIN (i.e. NEC)

- Pacifier.
- Careful handling.
- Supportive positioning.
- Skin to skin kangaroo care.
- Consider an IV morphine sulphate infusion or IV Paracetamol.
- Positive touch from parents.

10. LUMBAR PUNCTURE

- Pacifier + Breast milk (or sucrose if not available) 2 minutes prior to procedure
- Consider Infiltration of lignocaine.
- Supportive positioning.
- Containment holding.
- Wrapped tucked position (Ramsay and Jackson 2018)

11. RETINOPATHY SCREENING

- Pacifier + Breast milk (or sucrose if not available) 2 minutes prior to procedure
- Local anaesthesia eye drops.
- Swaddling/Containment holding.
- Supportive positioning.

12. INFANTS EXPERIENCING TOTAL BODY COOLING

- IV morphine sulphate is considered the most appropriate treatment and should be given to both ventilated and non-ventilated infants.
- See the IV morphine sulphate drug monograph for dosage.
- Consider intermittent boluses of morphine instead of a continuous infusion to avoid a build-up of drug levels and subsequent respiratory depression. Cooled infants have a lower metabolic rate.
- Non-nutritive sucking.
- Containment holding.
- Supportive positioning.
- Positive touch from parents.

APPENDIX 2

The Astrid Lindgren and Lund Children's Hospitals Pain and Stress Assessment Scale for Preterm and Sick New born Infants (ALPS-Neo)

	0	1	2
Facial Expression	Peaceful	Distressed expression, may grimace slightly	Distressed expression, may cry Chin drop
Breathing Pattern	Calm, effortless breathing	Slightly strained breathing. Breathing pauses	Strained breathing. Fast breathing. Apneoa
Tone of extremities	Normal tone	Varied tone	Tense or flaccid
Hand/Foot Activities	Relaxed	Slightly clenched. May try to grasp. Hand on face	Tightly clenched. Fingers/toes spread. Flaccid
Level of Activity	Calmly awake Calmly asleep	Occasional motor restlessness	Persistent motor restlessness Exhausted

(Lundqvis, Larsson and Kleberg, 2014)

Score	**Recommended Action**
0 to 2	No action
3 to 4	Try non-pharmacological interventions and evaluate impact
5 or more	Try non-pharmacological interventions and repeat assessment at 5 minute intervals
Continues at 5 or more	If assessments continue to be more than 5, consider pharmacological pain relief following consultation with a senior doctor

(FINE partnership, 2015)

APPENDIX 3

Manual for ALPS-Neo pain and stress assessment scale

An infant in a balanced state shows organized behavior, which corresponds to the scale's *score of 0* in the respective behavioral items. In the next step, *score 1,* the infant can show initial signs of pain/stress as well as signs of attempting to regain balance. The infant may put a hand over its face (defensive behavior), try to grasp something to hold onto or squirm (find a more comfortable position) and may partially or briefly succeed. An infant who is assessed at s*core 2* in any of the behavioral items is in an unbalanced state, affected by pain or stress, and needs, if the total sum is high, non-pharmacological individualized supportive care intervention and/or pharmacological treatment in order to regain balance/well-being.

A high total sum can mean that the infant is in a state of total imbalance, e.g. it has a serious lung condition in which the imbalance in the physiological subsystem spills over into the motor and state regulating subsystems [1, 2], or that the infant is in serious pain. The infant can also be assessed at a high level on the scale because of hunger, a wet nappy, uncomfortable position, etc. The first step in such cases is to carry out an individualized supportive care intervention. If the infant responds positively to this intervention – obtaining a low total score in a new assessment – then the infant was expressing signs of dissatisfaction/stress and the intervention can be considered as an assessment of the infant's consolability. If, on the contrary, the infant continues to obtain a high total score, it is most probably experiencing pain.

It is sufficient for the infant to show only one of the behaviors in each assessment cell for it to be attributed to the current score. When the infant shows signs from two different assessment cells, the higher value applies (for example when the infant shows signs corresponding to both score1 and 2, the infant will be attributed the score of 2). The assessment of hand/foot activity is difficult when the infant is tucked up. In that case, scores must be assessed for hand/foot activity in relation to the infant's other behaviors. If the infant is distressed or experiencing pain, the assessment of the other parameters will be high, so the risk of underestimating the score value is small.

Individualized supportive care intervention is always the primary alternative to optimize the infant's condition, as long as the infant does not have an obvious reason for pain, due to post-operative conditions, a vacuum extraction delivery or thoracic drainage, for example. When individualized supportive care intervention does not have sufficient effect, the infant needs pharmacological treatment in order to be free of pain. In order to ensure that the infant receives safe and adequate pharmacological treatment, the recommendations for considering this must follow an established algorithm, based on the pain score results. Below is an example of how this can be designed:

1) If the infant gets a total ALPS-Neo score of $\geq$3 but <5 the non-pharmacological individualized supportive care intervention is carried out and subsequently a follow-up assessment is carried out to evaluate the effect of the intervention.
2) If the infant gets a total score of $\geq$5 points the non-pharmacological individualized supportive care intervention is carried out and the assessment is repeated at 5-minute intervals. If the assessment continues to be $\geq$5 pharmacological pain relief should be considered in consultation with the physician in charge.

FACIAL EXPRESSION

0. ***Peaceful**:* Can look for something to suck on or is sucking
1. ***Distressed expression, may grimace slightly***: Furrow between the brows, squeezes eyes shut, may pull the corners of the mouth sideways
2. ***Distressed expression, may cry or chin drop**:* Tenses the facial muscles, screws face up or lies with open mouth, exhausted facial expression

BREATHING PATTERN

0. ***Calm effortless breathing**:* Breathes calmly with or without respiratory support
1. ***Slightly strained breathing, breathing pauses**:* Deviates somewhat from the infant's "basic pattern", e.g. slightly faster, slightly more irregular and/or indications of laboured breathing (light recession, nasal flaring, grunting). Can exhibit brief respiratory pauses
2. ***Strained breathing, fast breathing, apnoea**:* Increased respiratory effort with clear recession and nasal flaring, grunting, can exhibit long respiratory pauses. Can alternate between breathing deeply and superficially, possibly taking deeper breaths after a period of superficial breathing/apnoea

TONE OF EXTREMITIES

0. ***Normal tone**:* Neither tense nor flaccid
1. ***Varied tone**:* Here the infant starts to be affected and can then alternate between losing strength and thereby tone, being tense or regaining normal tone.
2. ***Tense or flaccid**:* If the infant is flaccid it lies heavily on the bedding

HAND/FOOT ACTIVITY

0. ***Relaxed**:* Can grasp something lightly, is neither tense nor limp
1. ***Slightly clenched, may try to grasp, hand on face**:* Places hand over the face as though to protect/shield itself
2 ***Tightly clenched, fingers/toes spread or flaccid***

LEVEL OF ACTIVITY

0. ***Calmly awake, calmly asleep**:* Calm and satisfied. The infant holds its arms and legs bent and still, close to the body with normal muscle tone
1. ***Occasional motor restlessness**:* Moves in an agitated way, stretches arms and legs. Can intermittently calm down by bringing the arms and legs in closer to the body
2. ***Persistent motor restlessness or exhausted**:* Stretches arms and legs, does not calm down, or looks tense and exhausted, does not have the energy to react

(Kleberg and Lundqvist, 2014)

Appendix 4

Checklist for Optimising Neurodevelopment for Neonatal Procedures

Heel Lances | Ultrasounds | Dressing changes | Extubation | Central lines
Cares | Cannulation | Venepuncture | Lumbar puncture | ROP screening

Pre-procedure:

- ☑ Aim for at least 50 minutes undisturbed
- ☑ Not woken from restful sleep or already upset
- ☑ Colostrum/Breast Milk/sucrose given prior to procedure
- ☑ Massage given prior to procedure

During procedure:

- ☑ "4 HANDS" (parent/carer available to assist)
- ☑ Reassuring and soothing voice during procedure
- ☑ Cradled/held securely to chest by parent/caregiver OR ☑ High support and boundaries in cot
- ☑ Dummy/breast offered to baby if unsettled
- ☑ Baby allowed time to recover from any instability
- ☑ Baby helped to grasp hands and/or feet together
- ☑ Reassurance with hands and soothing voice
- ☑ No or limited interruptions during the procedure

Environment:

- ☑ Off/dimmed lights and baby's eyes shielded
- ☑ Calm environment with little activity
- ☑ Hushed voice and only occasional muted sounds

Post-procedure:

- ☑ Baby left settled after procedure
- ☑ Dummy/breast offered to baby after procedure

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